Steven Buchanan

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Mathematics PhD candidate (geometric analysis and nonlinear PDE) with applied quantitative research experience in option-implied densities, distributional forecasting, statistical modeling and machine learning in Python. Targeting quantitative research roles in derivatives, risk and systematic strategy.

Education

Australian National University: PhD in Mathematics — Sept 2023 – est. Feb 2027
– Research in geometric analysis and nonlinear PDE
– Visiting PhD Researcher, University of Toronto, Aug 2025 – Jan 2026

University of Tennessee: PhD studies in Mathematics — Aug 2021 – Sept 2023
– transferred to ANU

Tennessee Technological University: M.S. in Mathematics — Aug 2019 – May 2021

Tennessee Technological University: B.S. in Mathematics — Aug 2015 – May 2019

Quantitative Research & Projects

Market-Implied Stock Price Densities – Erdős Institute certification project (Sep 2025)

- Built an end-to-end Python pipeline recovering risk-neutral terminal-price densities from sparse, noisy historical SPX option cross-sections via the Breeden–Litzenberger identity. Overview available on my Github.
- Compared two recovery methods: implied-volatility smoothing, and a shape-constrained call-price fit enforcing the monotonicity and convexity required by static no-arbitrage before numerical second differentiation.
- Backtested ~6,900 implied distributions against realized SPX settlements using CRPS; the constrained fit cut average relative CRPS from 0.0401 to 0.0345, a ~14% improvement.
- Mapped risk-neutral to real-world densities through a consumption-based pricing kernel, estimating the CRRA risk-aversion parameter that minimized historical forecast error.

Global material stock database – Machine learning research team at ANU (current)

- Contribute quantitative and computational work to a team assembling a harmonized global database of building-material stocks from heterogeneous datasets.
- Evaluate candidate modeling approaches for classification and regression of building types and material stock use.
- Maintain reproducible analytical workflows and translate technical modeling issues for a cross-disciplinary group.

Academic Research

Deformation of Surfaces by Functions of Curvature – Doctoral Research

- Research nonlinear geometric evolution equations generalizing Ricci/Yamabe flow, using techniques from differential geometry, nonlinear parabolic PDEs, and asymptotic analysis.
- Analyze existence, singularity formation, and long-time behavior of curvature-driven systems; establish estimates leading to convergence of compact non-negatively curved surfaces toward round geometries under rescaling.
- Develop and apply evolution equations, maximum principles, differential Harnack inequalities, entropy/monotonicity methods, and scaling arguments to control nonlinear PDE solutions.

Asymptotic Analysis of Arithmetic Functions – Master's Thesis

- Derived asymptotic formulae for sums of classes of divisor functions over arithmetic progressions using methods from analytic number theory.

Probabilistic modeling of divisibility patterns in integers – Undergraduate research project

- Conducted asymptotic and probabilistic analysis of divisibility relations to quantify how often a generalized version of Euclid's lemma holds under random integer sampling.
- Published at: S. W. Buchanan and A. J. Hetzel, "On the Success of Seriously Mishandling Euclid's Lemma," *American Mathematical Monthly* **127** (2020), no. 5, 401–410.

Skills

Programming & data analysis: Python (NumPy, SciPy, pandas, PyTorch, Matplotlib, Jupyter), SQL, TypeScript, Git, Linux command line, LaTeX

Quantitative & financial: risk-neutral and option-implied densities, no-arbitrage constraints on option prices, pricing kernels and CRRA utility, probabilistic forecasting and CRPS, backtesting and model validation, Monte Carlo simulation, time-series analysis, stochastic processes, numerical optimization

Statistics & ML: regression, random forests, gradient boosting, hyperparameter tuning, model development and validation, data cleaning and integration, reproducible pipelines

Mathematics: nonlinear parabolic PDE, differential geometry, probability and statistics, asymptotic analysis, numerical analysis, linear algebra, analytic number theory

Additional

Teaching: Instructor and teaching assistant for differential geometry, differential equations, linear algebra, discrete mathematics, and college algebra at ANU, Shandong University, University of Tennessee and Tennessee Tech. Designed lectures, assessments and problem-solving tutorials for undergraduate students.

Awards: HDR Fee Remission Merit Scholarship, International PhD scholarship – Australian National University; F. M. Dryzer Math Fellowship – University of Tennessee

Work rights: U.S. citizen; currently in Australia on a student visa with working rights